Cash Flow Intelligence Statement — V1.0

Platform: Langflow v0.5.0.post2 | **LLM:** OpenAI o3-mini | **Last updated:** 2026-02-01

What it does

Takes a bank statement (PDF or CSV) and a loan application JSON, then automatically tags every transaction, detects hidden EMIs, and returns an ACCEPT / REJECT decision with supporting metrics (RMI, FOIR, risk-spend ratio, EMI-day balance).

Inputs

Input	Where	Format
Bank statement	PDF upload → *Generic Bank Statement Extractor*, **or** CSV upload → *CSV File Reader*	PDF / CSV
Loan application	Chat prompt → *Chat Input*	JSON (see below)
API key	*Language Model* component	OpenAI key

Loan JSON schema:

```json
{
  "employment_type": "SALARIED",
  "requested_loan_amount": 5000000,
  "tenure_months": 180,
  "proposed_emi_estimated": 45000
}
```

`proposed_emi_estimated` is optional — if omitted, EMI is calculated at 12 % p.a.

Pipeline (execution order)

1 — Ingestion (two parallel paths, one winner)

#	Component	Does
1a	**Generic Bank Statement Extractor**	OCR-extracts a PDF into CSV text using geometry + balance-based Debit/Credit classification. Validates the document is actually a bank statement before extracting.
1b	**Bank Statement Data Parser**	Parses the CSV text into a list of row-level Data objects.
1c	**Transaction Table Output**	Wraps the row list into a single Data object.
1d	**Type Converter (PDF path)**	Data → DataFrame.
2a	**CSV File Reader**	Reads an uploaded CSV into a Data object.
2b	**Type Converter (CSV path)**	Data → DataFrame.
3	**Router Component**	Picks whichever path produced valid data (PDF wins if both did). Flattens Debit/Credit columns into a single `amount` + `type` column and coerces numerics.

2 — Tagging

#	Component	Does
4	**Language Model (o3-mini)**	Supplies the LLM; system prompt defines the five tag categories.
5	**Batch Run**	Sends every row to the LLM individually; writes the tag into `model_response`.
6	**Hidden EMI Detector**	Deterministic post-pass that (a) sanitises LLM mistakes (e.g. debits tagged as income), (b) strips all LLM-applied `HiddenEMITxn` tags, (c) re-detects them properly by scanning for recurring debits across ≥ 3 of 6 months at ±5 % tolerance, and (d) unifies inconsistent tags on the same payee via majority vote.
7	**DataFrame Operations**	Renames `model_response` → `Bucket`, drops the `batch_index` helper column.

3 — Decision

#	Component	Does
8	**Chat Input**	Receives the loan JSON from the user.
9	**JSON Input**	Parses and validates the JSON into a Data object.
10	**Loan Eligibility Checker**	Calculates metrics and applies decision rules (see below). Outputs ACCEPT or REJECT with metadata.

4 — Output

#	Component	Does
11	**Chat Output**	Displays the decision JSON in the playground.

Transaction Tags

Tag	Meaning	Assigned by
IncomeTxn	Salary / business inflow credits	LLM
DetectedEMITxn	Loan EMI debits (narration contains EMI / NACH / LOAN etc.)	LLM
HiddenEMITxn	Shadow-loan debits (same amount repeats ≥ 3 / 6 months, lender-like narration)	Hidden EMI Detector
RiskSpendTxn	Risky debits (gaming, betting, crypto, liquor, etc.)	LLM
ReturnTxn	Bounces, returns, penalties	LLM
None	No category matched	LLM / sanitiser

Investment narrations (SIP, mutual fund, broking platforms) are **never** tagged as EMI — excluded at both the LLM prompt level and the deterministic sanitiser.

Decision Rules

The Eligibility Checker rejects if **any** of the following is true:

Rule	Condition
Insufficient history	Fewer than 6 months of statement data
High FOIR	(Existing EMI + Proposed EMI) / RMI > 60 %
Low EMI-day balance	Balance on the 5th of the month is below the proposed EMI in ≥ 2 months
Low income	RMI < 1.5 × Proposed EMI

Output metrics: Real Monthly Income (RMI), Real FOIR, Risk Spend Ratio, Avg Monthly Balance, Existing EMI total, Proposed EMI, number of months analysed.

Output Samples

The Chat Output renders the JSON returned by the Loan Eligibility Checker. Two representative cases are shown below.

ACCEPT

The applicant's income comfortably covers existing and proposed obligations; balance on EMI day is adequate in all months.

```json
{
  "decision": "ACCEPT",
  "metadata": {
    "Real Monthly Income": 152000.00,
    "Real FOIR": 0.3782,
    "Risk Spend Ratio": 0.0421,
    "reject_reasons": null,
    "num_months": 6,
    "AvgMonthlyBalance": 87400.50,
    "ExistingEMI": 12500.00,
    "EMI_proposed": 45000.00
  }
}
```

Metric	Value	Threshold	Status
Months of data	6	≥ 6	✓
FOIR	37.82 %	≤ 60 %	✓
EMI-day balance shortfall	0 months	< 2 months	✓
Income vs EMI (1.5×)	152 k ≥ 67.5 k	RMI ≥ 1.5 × EMI	✓

REJECT — High FOIR

The applicant already carries a large existing-EMI burden (including one hidden EMI detected by the deterministic scanner), pushing the combined FOIR above 60 %.

```json
{
  "decision": "REJECT",
  "metadata": {
    "Real Monthly Income": 98000.00,
    "Real FOIR": 0.7347,
    "Risk Spend Ratio": 0.1130,
    "reject_reasons": [
      "High FOIR: 73.47%"
    ],
    "num_months": 7,
    "AvgMonthlyBalance": 31200.75,
    "ExistingEMI": 26950.00,
    "EMI_proposed": 45000.00
  }
}
```

Metric	Value	Threshold	Status
Months of data	7	≥ 6	✓
FOIR	73.47 %	≤ 60 %	**✗ reject**
EMI-day balance shortfall	1 month	< 2 months	✓
Income vs EMI (1.5×)	98 k ≥ 67.5 k	RMI ≥ 1.5 × EMI	✓

> `ExistingEMI` here is the monthly average of all `DetectedEMITxn` + `HiddenEMITxn` debits combined. The hidden portion (e.g. a recurring 8 950 ₹ debit to a lending-gateway narration across 4 months) is added only after the deterministic scanner runs — it would be invisible to the LLM alone.

Key design notes

- **Hidden EMI detection is fully deterministic.** The LLM sees rows one at a time and cannot detect cross-month recurrence. The Hidden EMI Detector component strips all LLM `HiddenEMITxn` tags and re-derives them using the full dataset.
- **Fuzzy payee matching.** The sanitiser normalises narrations (strips UPI/NACH prefixes, ref numbers, casing) and applies a majority-vote per payee group to eliminate tag inconsistency across runs.
- **PDF extraction uses OCR + financial validation.** Debit vs Credit is determined first by balance-change logic, then by column-position detection when no previous balance is available. A pre-flight check rejects documents that don't look like bank statements.
- **Router priority:** If both PDF and CSV paths produce valid data, the PDF path is used.